

Student Name: \_\_\_\_\_

Class Name: **College Algebra FA26 - 014 MW 2:30**

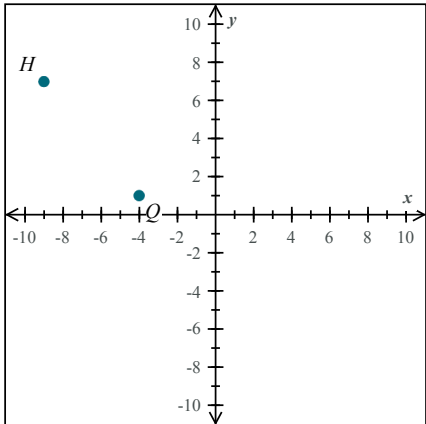
Number of Questions: **12**

Instructor Name: **Mamani Velasco, Ruben**

Question 1 of 12

Calculate the distance between the points  $H = (-9, 7)$  and  $Q = (-4, 1)$  in the coordinate plane.

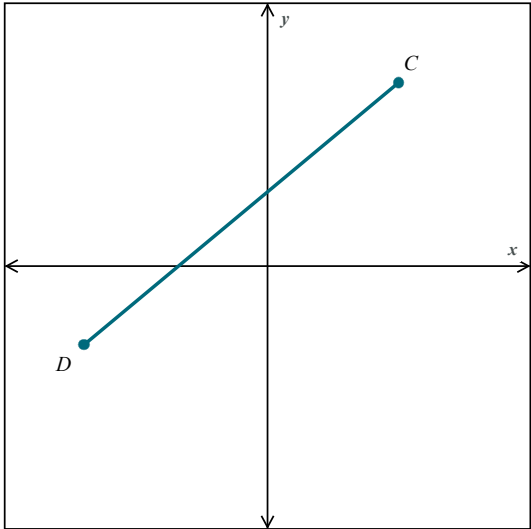
Give an exact answer (not a decimal approximation).



Distance: \_\_\_\_\_

Question 2 of 12

Find the midpoint  $M$  of the line segment joining the points  $C = (5, 7)$  and  $D = (-7, -3)$ .



Question 3 of 12

Find the  $x$ -intercept(s) and  $y$ -intercept(s) of the graph of the following.

$$x^2 + y - 25 = 0$$

If there is more than one answer, separate them with commas.

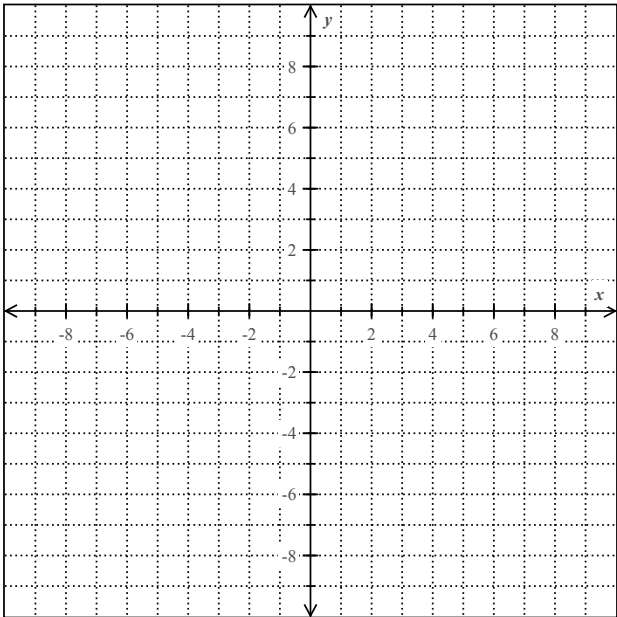
Question 4 of 12

The equation of a circle is given below. Identify the radius and center. Then graph the circle.

$$(x - 2)^2 + (y - 5)^2 = 16$$

Radius: \_\_\_\_\_

Center: (\_\_\_\_\_, \_\_\_\_\_)



Question 5 of 12

The equation of a circle is given below. Identify the radius and the center. Then graph the circle.

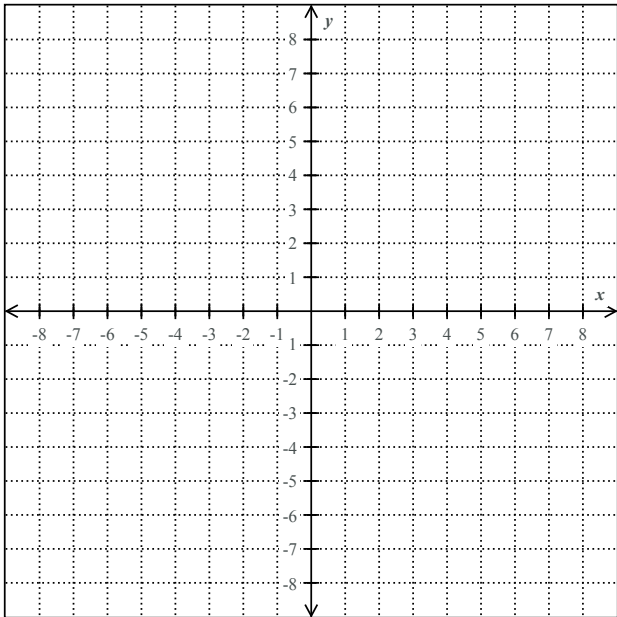
$$x^2 + y^2 + 10x + 4y = -20$$

(a) Identify the center and the radius of the circle.

Radius: \_\_\_\_\_

Center: ( \_\_\_\_\_ , \_\_\_\_\_ )

(b) Graph the circle.



Question 6 of 12

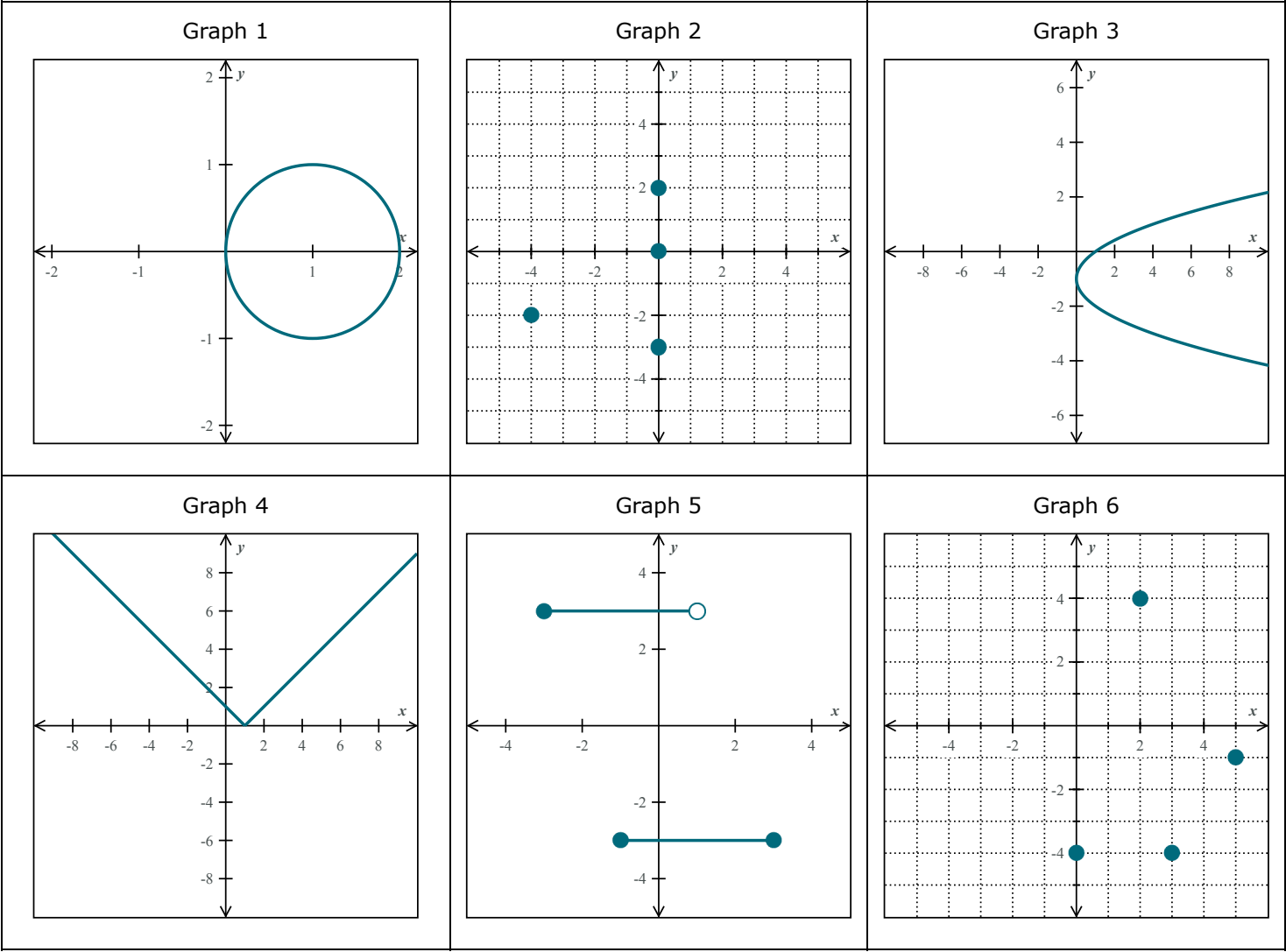
Write an equation of the circle with center (5, 9) and diameter 12.

Question 7 of 12

Find an equation of the circle that has center (6, -2) and passes through (2, -6).

Question 8 of 12

Six graphs are shown below. Click on the tabs to see them.



Check the box corresponding to each graph that represents a function.

- ☐ Graph 1
- ☐ Graph 2
- ☐ Graph 3
- ☐ Graph 4
- ☐ Graph 5
- ☐ Graph 6
- ☐ None of the graphs

Question 9 of 12

The function  $h$  is defined as follows.

$$h(x) = \frac{x - 9}{x + 8}$$

Find  $h(-2)$ .

Simplify your answer as much as possible.

Question 10 of 12

Find the domain of the function.

$$g(x)=\sqrt{-x}-3$$

Write your answer using interval notation.

Question 11 of 12

Leila received a \$30.00 gift card for a photo center. She used it to buy prints that cost 9 cents each. The remaining balance,  $B$  (in dollars), on the card after buying  $x$  prints is given by the following function.

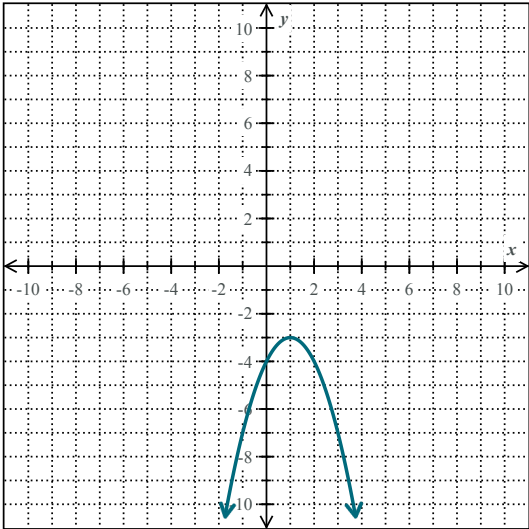
$$B(x)=30.00-0.09x$$

What is the remaining balance on the card if Leila bought 50 prints?

dollars

Question 12 of 12

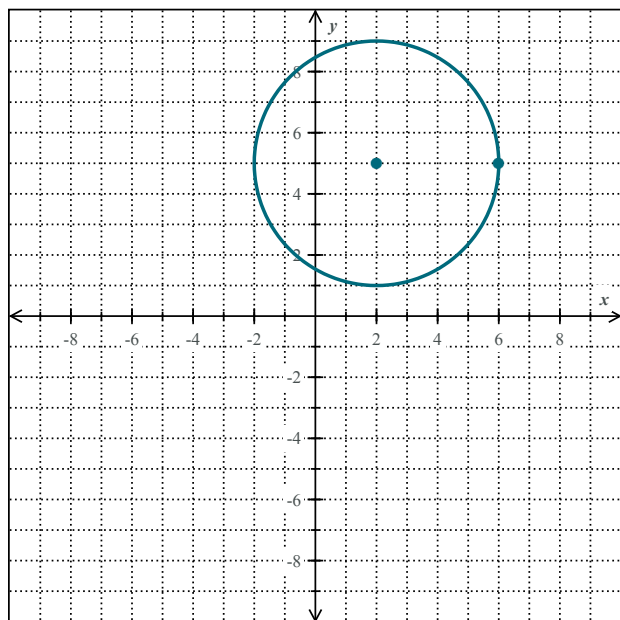
The graph of a quadratic function with vertex  $(1, -3)$  is shown in the figure below. Find the domain and the range.



Write the domain and range using interval notation.

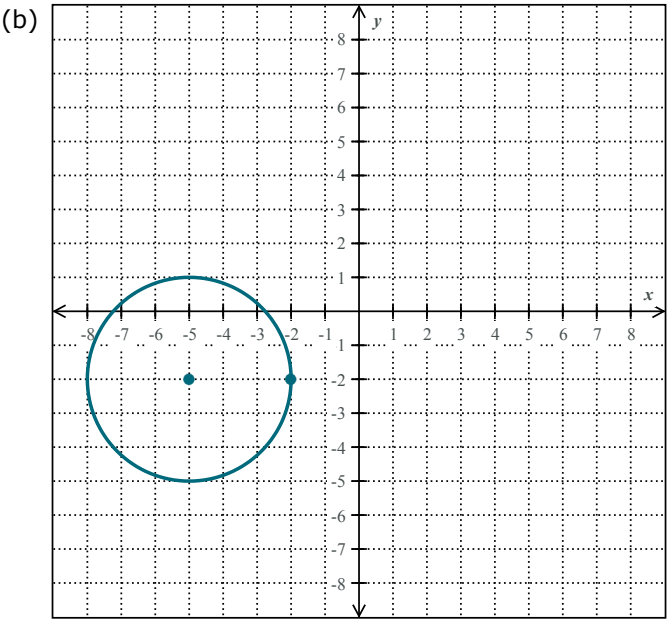
(a)domain= \_\_\_\_\_

(b)range= \_\_\_\_\_

Class Name: **College Algebra FA26 - 014 MW 2:30**Number of Questions: **12****Question 1 of 12**Distance:  $\sqrt{61}$ **Question 2 of 12** $M = (-1, 2)$ **Question 3 of 12** $x$ -intercept(s):  $5, -5$  $y$ -intercept(s):  $25$ **Question 4 of 12**Radius:  $4$ Center:  $(2, 5)$ 

Question 5 of 12

- (a) Radius: 3  
Center:  $(-5, -2)$



Question 6 of 12

$$(x - 5)^2 + (y - 9)^2 = 36$$

Question 7 of 12

$$(x - 6)^2 + (y + 2)^2 = 32$$

Question 8 of 12

- ☐ Graph 1    ☐ Graph 2    ☐ Graph 3    ☒ Graph 4    ☐ Graph 5    ☒ Graph 6    ☐ None of the graphs

Question 9 of 12

$$h(-2) = -\frac{11}{6}$$

Question 10 of 12

$$(-\infty, 0]$$

Question 11 of 12

25.50 dollars

**Question 12 of 12**

(a) domain:  $(-\infty, \infty)$

(b) range:  $(-\infty, -3]$